

PRV Stations & Booster Pumps: Compliance, SOP & Optimization Guide

Purpose:

PRVs regulate pressure for downstream safety and equipment life; booster pumps maintain water flow in low-pressure zones.

Why It's Ignored:

Seen as 'set once and forget'; PRV drift or booster cavitation often undetected until pressure loss or failure.

Must-Have Docs:

- Pressure setpoint logs
- Valve rebuild records
- Pump curve validation
- Preventive maintenance SOPs

Compliance Officials:

IPC, ASME B31, Local Plumbing Code, OSHA

PPE Required:

Gloves, goggles, hearing protection during testing or bleed-down

Installation Practices:

- Install PRVs with gauge taps and isolation valves
- Mount booster pumps on anti-vibration pads
- Include expansion tanks and relief valves where required

Must Checks:

- Downstream pressure stability
- PRV weep/drip or fail-open
- Pump runtime and amp draw
- Valve noise or chattering
- System air bleed or surge suppression

Regulations & Codes:

IPC, OSHA, ASME B31, Manufacturer specifications